Conduction vs Insulation by RealQM

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Abstract

This article shows what RealQM, as a new alternative to textbook quantum mechanics, offers as explanation of the difference between conductors and insulators. RealQM represents an N-electron system by N non-negative unit charge densities in ordinary three-dimensional space, one per electron, meeting at free boundaries and minimising the total energy — Coulomb interaction plus kinetic energy, with no other force and nothing fitted. It is a computable, parameter-free model at polynomial cost.

Computationally the difficulty is one of scale. Atomic energies are tens of electron-volts; semiconductor activation is 50–500 meV; metallic conduction sets in near 20 meV. Against the total energy of a system large enough for the question to be posed those are 10^{-3}, 10^{-5} and 10^{-6}, so whether a material conducts is decided by a difference five or six orders of magnitude below the number a calculation returns. The same span falls on the model: a description adequate for binding, where a per cent error is invisible, can be hopeless for conduction, where a per cent of the total is a hundred times the quantity in question.

Two models are measured. A **chain of atoms**, the valence electron occupying its own kernel's region, gives a corrugation of order an electron-volt for the electron pattern to slide through, and an extensive polarisability $\alpha = 27.4N$: an insulator. A **cable of ions surrounded by electrons**, the carriers standing off cores that screen the kernels, gives ≈240 meV — the semiconductor range, where transport is thermally activated hopping. Raising the coefficient of the kinetic energy from the $\frac{1}{2}$ of atomic standardisation to 1, which compensates for a free interface that resists neither compression nor modulation, gives ≈24 meV: the pinning no longer holds and charge is free to move.

The corrugation is the activation energy E_a in the hopping conductivity $\sigma = (ne^2a^2\nu/kT)\exp(-E_a/kT)$. Closing the prefactor with an attempt frequency $\nu = (1/2\pi)\sqrt{K/M}$ built from the curvature of the same landscape and an ionic mass gives $\nu \approx 10^{12}$–10^{13} Hz, a lattice frequency, and room-temperature conductivities of 2×10^{-12}, 6.5 and 8.8×10^3 $\mathrm{S\,m^{-1}}$ for the three cases — glass, germanium and a poor metal, in that order, with nothing fitted. Only E_a is measured here; the mass and the carrier density are inputs, since RealQM carries no masses. Metallic conduction proper is different: its rate needs a relaxation time from electron–phonon scattering, and that stays outside.

1 The framework, and what is asked of it

RealQM represents an N-electron system by N non-negative densities ψ_i, each carrying unit charge on its own domain Ω_i, the domains non-overlapping and their interfaces free to move to where the energy is least. The energy minimised is

$$E[\{\psi_i\},\{\Omega_i\}] = \sum_i \int_{\Omega_i}\Big(\tfrac{1}{2}|\nabla\psi_i|^2 - \sum_a \frac{Z_a}{|x-x_a|}\psi_i^2\Big)dx + \tfrac{1}{2}\sum_{i\neq j}\iint \frac{\psi_i^2(x)\psi_j^2(y)}{|x-y|}\,dx\,dy, \qquad (1)$$

subject to $\int_{\Omega_i} \psi_i^2 = 1$. There is no wavefunction in $3N$ dimensions, no antisymmetry and no self-interaction; the role played by the exclusion principle is played by the requirement that domains do not overlap. The minimisation runs over the partition as well as over the densities, so the domain edges are unknowns of the problem. Everything is Coulomb interaction and kinetic energy, with nothing fitted, and the cost is polynomial in N rather than exponential.

Every previous test of this construction — ionisation energies across the periodic table, molecular binding, hydrogen bonds and solvation, nuclear binding by dual Coulomb confinement — has been on *bound* matter, where an attractor is always present and the charge is where the attractor is. Conduction is the first question that asks about charge which is not where an attractor holds it, and the framework has no separate machinery for it: the same energy, the same domains, in a different arrangement of the same two kinds of charge.

What conduction is normally computed with. There is no single standard theory of conduction to set beside this one. There is a stack of them, each excellent within its regime and each carrying quantities supplied from measurement. Metal is separated from insulator by a gap at the Fermi level, and the gap comes from a chosen exchange–correlation functional which moves it by tens of per cent, repaired by hybrid mixing or a GW self-energy. Transport is then a second model on top of the first: a Boltzmann equation closed by a relaxation time, with an effective mass and a deformation potential for the semiconductor; a transfer integral and a reorganisation energy for a hopping conductor; a localisation length and a density of states at the Fermi level for variable-range hopping in a disordered one; a Hubbard U where the band picture fails outright and the material insulates by correlation instead. The point of the contrast is not accuracy — these are accurate, and are what one should use to predict a number for a given material. It is that they are several models with parameters, addressing the three regimes separately, whereas what follows is one energy with none, in which the regimes differ by how the charge is arranged.

Two arrangements, both ordinary matter. The first is a **chain of atoms**: one kernel per site with its valence electron occupying the same region, which is what a hydrogen chain is and what an alkali is to a first approximation. The second is a **cable of ions surrounded by electrons**: kernels screened by their own core electrons, with the valence charge standing off outside them, which is how a wire is built. The question is what each gives.

2 What is measured

A domain carries one unit of charge by constraint, so no charge passes between sites. What can move is the *partition*: the boundaries are redrawn, and the pattern of domains advances through the lattice. When the pattern has advanced by one full spacing, every electron has taken over its neighbour's kernel and the configuration is restored — identical, since the electrons are identical and the energy depends on the partition and not on labels. Charge has been transported and nothing material has traversed the sample, which is the relation a dislocation bears to slip.

The quantity that decides whether this happens is the energy along that path. Write $E(d)$ for the energy with the partition displaced by a fraction d of a spacing. It is periodic, $E(d+1) = E(d)$, and its amplitude

$$V_0 = \max_d E(d) - \min_d E(d) \qquad (2)$$

is the corrugation: the barrier the partition must cross to advance one site. A field tilts this curve by $-NeFd$, so the pattern runs when the tilt exceeds the steepest slope of $E(d)$, and otherwise settles at a displacement and stops — bounded polarisation, no current.

Ends, and how to remove them. A finite chain has ends, and they dominate: terminal domains bind an electron-volt harder than interior ones, so a corrugation read from total energies is reading the ends. Displacing only the middle M carriers repairs this. The energy difference is then M times the bulk cost plus two junction costs, so differencing two window sizes cancels the junctions exactly, and no terminal domain moves in either configuration:

$$\Delta E(M) = M\,\varepsilon_{\text{bulk}} + 2\varepsilon_{\text{junc}}, \qquad \varepsilon_{\text{bulk}} = \frac{\Delta E(M_2) - \Delta E(M_1)}{M_2 - M_1}. \tag{3}$$

The decomposition asserts linearity in M, which a third window tests rather than fits.

3 Model I: a chain of atoms

Seven model atoms in a row, one valence electron each, the electron occupying its kernel's region. Two measurements.

The polarisability is extensive. With the partition held fixed and a uniform field applied, the induced dipole gives α, and repeating at three chain lengths gives its scaling: $\alpha = 27.4N$, flat to 1% per atom. That is the signature of charge fixed per site. A metal's polarisability grows faster than the number of atoms, because charge redistributes over the whole sample; here each domain carries its unit and cannot give it up, so the only responses are the density deforming inside a fixed domain and the boundaries displacing. Both are per-atom constants, and the sum is exactly proportional to N. The one escape is a vanishing corrugation, which would let the partition slide freely and the response grow without bound.

The corrugation does not vanish. It is of order an electron-volt per electron, and it stays there: repeating the measurement across lattice spacings from 3.5 to 8.0 a_0 gives a minimum near the equilibrium spacing and a rise on both sides, with no spacing approaching zero and no flip of the registry preference. Compressed, the cores crowd the electron; expanded, sharing a kernel costs more than owning one.

An electron-volt is forty times thermal energy at room temperature and lies above the 10–500 meV of the activated conductors this framework might otherwise describe. On this model the answer is unambiguous: an insulator, and not a marginal one.

4 Model II: a cable of ions surrounded by electrons

The chain places the carrier where the attractor is. Conducting matter does not: the core electrons occupy the region around each nucleus and the conduction charge stands outside them. Representing that needs two populations — cores held inside a radius, carriers outside it, with only the carriers free to slide — and it is the one arrangement that gives a carrier a standoff without violating the tiling, because the inner region is genuinely owned rather than left empty.

Sixteen sites, kernels of charge +3 on the axis, a closed core of two electrons at each, and one valence electron per site outside them: neutral, with no pseudopotential anywhere, so the exclusion of the carrier from the core region is by non-overlap rather than by an imposed condition. The corrugation, measured by (3) with windows of four and eight carriers and checked against a third at twelve:

$$\varepsilon_{\text{bulk}} \approx 240 \text{ meV per carrier.}$$

That is inside the 10–500 meV band of the doped semiconductors, organic conductors and polaronic oxides — the class of materials whose transport is thermally activated hopping, and the class this framework has a claim on. The same interaction, the same energy, the same domains: only the arrangement of the charge has changed, and the barrier has fallen by a factor of four.

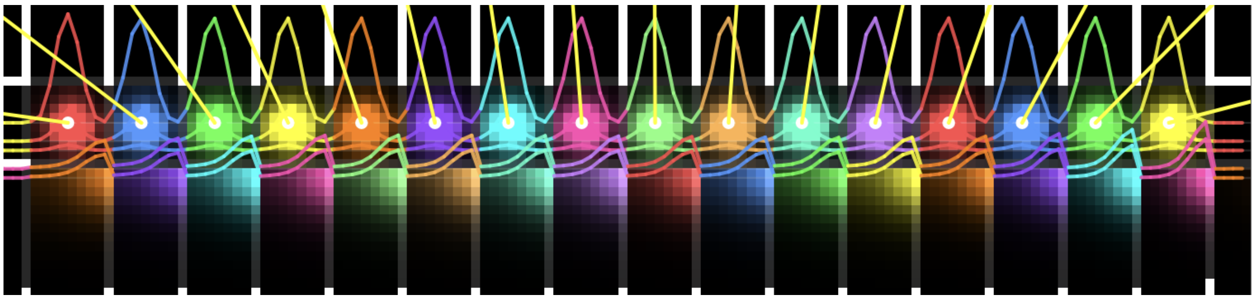

Figure 1: The cable, relaxed. Sixteen sites along the axis, each carrying a kernel of charge +3, a core domain of two electrons confined within $c_r = 2.4\,a_0$, and one valence electron outside it. Colour distinguishes domains: the bright lobes on the axis are the cores, the fainter blocks beneath them the valence charge standing off. White verticals mark the free boundaries between neighbouring carriers, and the coloured curves are density profiles along a line through the axis; the yellow segments are net forces on the kernels. Every boundary shown is an unknown of the minimisation, not an imposed partition. Displacing the middle M of these carriers by half a spacing, at fixed cores and kernels, is the measurement of §2.

5 What sets the pinning

The barrier is not set by the strength of the ion lattice but by whether the carriers bunch into its wells. A carrier distribution that resisted being modulated would ride over the lattice; one that does not resist locks to it.

RealQM's free interface resists very little, and this is internal to the construction. A flat density is admissible on a free boundary at no cost — a constant ψ_i satisfies the normalisation, has $\nabla\psi_i = 0$, and minimises the electrostatic energy against a neutralising background — so redrawing an interface between domains of equal density costs nothing. The partition therefore offers no resistance to uniform compression, and the only penalty on a *modulated* density is the gradient term in (1), carrying the coefficient $\frac{1}{2}$.

That coefficient is a standardisation, not a constant of the theory. It is set by calibrating the framework against atomic spectra, hydrogen first among them — on a bound electron in a single potential well. An extended arrangement is a different object, as an effective mass in a solid differs from the free-electron mass without anyone regarding it as tampering, and the coefficient governing how a carrier lattice resists modulation by an ion lattice has no obligation to equal the one an atom's spectrum sets. Raising it is therefore a question rather than a fitted parameter: what stiffness does an extended arrangement carry?

Raised from $\frac{1}{2}$ to 1, with the ions, the standoff, the geometry and the mesh unchanged:

$$\varepsilon_{\text{bulk}} \approx 24 \text{ meV per carrier},$$

a fall by a factor of eleven, to thermal energy at room temperature. The pinning no longer holds and the charge is free to move — not as a metal, since non-overlap leaves no extended states to fill into bands, but as a charge lattice sliding collectively over the ions.

model	corrugation per carrier	regime
chain of atoms: carrier on its own kernel	$\sim$1 eV	insulator
cable of ions: carriers standing off the cores	$\approx$240 meV	semiconductor: activated hopping
the same cable, carriers stiffened	$\approx$24 meV	metallic onset: unpinned

6 What this gives, and what it does not

In the activated regime the corrugation is the activation energy, and the conductivity of a hopping conductor is

$$\sigma = \frac{ne^2a^2\nu}{kT}\, e^{-E_a/kT}, \qquad \nu = \frac{1}{2\pi}\sqrt{\frac{K}{M}}, \tag{4}$$

a the hop distance, n the carrier density, ν the attempt frequency — the carrier sits in a well and tries once per vibrational period — and K the curvature of that well. The measurements supply E_a, which decides the temperature dependence and the ratio between regimes. The prefactor needs K, M and n, and it is worth being exact about where each comes from.

K is the curvature of the landscape whose amplitude was measured. Taking the periodic curve to be $E(u) = \frac{1}{2}V_0\big[1 - \cos(2\pi u/a)\big]$ gives $K = 2\pi^2 V_0/a^2$, which for $V_0 = 240$ meV at $a = 4.5\,a_0$ is 0.84 eV Å^{-2}. This is the one step that assumes a shape rather than measuring one: the windowed slide was run at the two endpoints $d = 0$ and $d = \frac{1}{2}$, which fix the amplitude and not the curvature. A scan at small d would replace the assumption, and until it is run the sinusoid stands in for it.

M is not a RealQM quantity at all. The framework carries no masses: the coefficient of $|\nabla\psi_i|^2$ is a geometric scale fixed by atomic standardisation, not $\hbar^2/2m$, which is precisely why §5 could raise it. So M enters from outside as the mass of whatever moves — and what moves here is the site, since the charge does not pass between sites but the pattern of domains advances through the lattice. M is therefore a lattice mass, and ν comes out a phonon frequency. This is a genuine seam: RealQM fixes the exponent, and one number for the prefactor is imported.

$n = 1/a^3$ takes the cable's spacing to hold in three dimensions, one carrier per site. The cable is a single line of sites, so this is a stated convention for converting a per-carrier barrier into a bulk conductivity, not something the calculation contains.

With $M = 28\,u$ and $T = 300$ K:

	E_a	ν (Hz)	σ (S/m)	comparable to
chain of atoms	$\sim$1 eV	5.5×10^{12}	2×10^{-12}	glass, 10^{-11}–10^{-15}
cable of ions	240 meV	2.7×10^{12}	6.5	intrinsic Ge, 2; lightly doped Si
cable, stiffened	24 meV	8.5×10^{11}	8.8×10^{3}	a poor metal; Cu is 6×10^{7}

The attempt frequencies are 10^{12}–10^{13} Hz, which is where lattice vibrations live; nothing was arranged to make that so, and it is the check that the mass entering ν is the right kind of mass. The conductivities span fifteen orders of magnitude and fall, in order, in the insulating, semiconducting and metallic-onset ranges. Between the two cable rows the factor is 1.3×10^3: the exponential alone gives 4×10^3, reduced because $\nu \propto \sqrt{V_0}$ falls as the barrier does.

These are order-of-magnitude statements and cannot be more. The ± 30 meV on E_a is already a factor 3.2 either way in σ at room temperature, the sinusoid is an assumption, and M and n are inputs. What the framework delivers is the exponent; that the resulting numbers land in the three named regimes, in the right order, with nothing fitted, is the result.

The kink, and why the table is a lower bound. Every barrier above is a rigid slide: all carriers displaced together, so the cost is N times a per-carrier barrier and the whole sample must climb at once. Real conductors do not do this. The cheap path is a kink — one carrier advancing while the rest hold still, the domains ahead of it compressed and those behind expanded — a localised defect whose energy does not grow with the length of the chain and which, once formed, moves by shifting its centre through the lattice. This is the relation a dislocation bears to sliding a whole crystal plane, and it is the same object a charge-density wave transports charge with.

The carriers-on-a-substrate system of §5 is a Frenkel–Kontorova chain: a harmonic coupling κ between neighbouring carriers on the periodic potential of curvature K measured here. Two consequences follow, and both bear on the numbers above. First, the barrier changes character. A kink of width $w = \sqrt{\kappa/K}$ sites crosses a Peierls–Nabarro barrier E_{PN} which falls steeply — exponentially, in the continuum limit — as w grows, so a wide kink slides almost freely over a substrate that pins a rigid chain hard. The conductivity is then governed by $E_f + E_{\mathrm{PN}}$, the cost of creating a kink plus the cost of moving it, in place of E_a; since $E_{\mathrm{PN}} \ll V_0$ the exponent falls and every σ in the table is an underestimate, by a factor this article does not determine.

Second, and to the question of a clock: the kink is better supplied with one than the rigid slide is. The same two constants give the carrier lattice a sound speed $c_0 = a\sqrt{\kappa/M}$, which is the kink's limiting velocity, and a kink mass set by κ, K and M — so the rate of kink transport is fixed by κ, K, E_f, E_{PN} and M, of which only M comes from outside. The width w in particular is a pure ratio of two curvatures and therefore wholly a RealQM quantity. Nothing new in kind is needed; what is needed is κ, which is the curvature of the energy against a *non-uniform* displacement of the carriers, and E_{PN}, which must be inferred from κ and K rather than read off a slide. Neither is measured here.

Metallic conduction is different and is outside. There $\sigma = ne^2\tau/m$, with τ a momentum relaxation time set by electron–phonon scattering, which requires a current as an object, electron dynamics, and irreversibility. A minimisation over real densities has none of the three. Non-overlap also leaves no extended states to fill into bands, so band conduction is absent in kind and not by degree, and no choice of stiffness produces it.

7 Status

The corrugations above are rigid-slide values, and as §6 sets out they are upper bounds: the kink is the cheap path and its barrier is not measured here. The states carry Euler–Lagrange residuals of 2–3 $\times 10^{-2}$, against 4×10^{-3} for the chain, so they are not fully relaxed; the stiffened runs use a halved timestep and are less relaxed than the others; and the factor of eleven rests on two values of the coefficient rather than a curve. The chain's corrugation depends on a pseudopotential exclusion rule that is not settled: excluding a domain from its own core alone under-excludes, excluding from every core over-confines, and the two bracket the answer rather than fixing it. What does not depend on any of this is the instrument. Equation (3) cancels ends and junctions by construction and tests itself against a third window, and the controls throughout are of the kind that can fail: translation by one spacing must return the same energy to the last digit, and domains related by a symmetry must have equal eigenvalues.

8 Conclusion

Asked whether it distinguishes a conductor from an insulator, RealQM answers differently for two arrangements of the same charge. Where the valence electron occupies its own kernel's region it insulates, with a corrugation of an electron-volt and a polarisability extensive in the number of electrons. Where the carriers stand off ion cores — the arrangement conducting matter actually has — the barrier falls into the range of the activated conductors. And the barrier is set not by the lattice but by how strongly the carrier distribution resists being modulated by it, which is a property of the interface condition rather than of the interaction: a free interface resists nothing, and supplying the missing stiffness removes the pinning.

What the framework delivers is the activation energy. Closing the hopping prefactor with an attempt frequency built from that landscape and an ionic mass then puts the three arrangements at 10^{-12}, 10^{0} and 10^{4} $\mathrm{S\,m^{-1}}$ — glass, germanium, a poor metal — with the attempt frequencies coming out at lattice values without being asked to. The mass is imported, since RealQM carries none; the carrier density is a convention; the curvature assumes a shape; and all three numbers are upper bounds on the barrier and hence lower bounds on σ, because charge moves by kinks and not by rigid slides. What the framework does not deliver at all is metallic conduction proper, whose rate is a relaxation time from electron–phonon scattering: that needs a current as an object and irreversible dynamics, and a minimisation over real densities has neither.

A Computational details

All calculations use the RealQM solver `molecule.js` driven by `chain_slide.html` (WebGPU, single GPU). Equation (1) is minimised by imaginary-time relaxation of the densities on a uniform Cartesian grid, with the partition specified by a cell-ownership map. For every corrugation the partition is *frozen*: the energy of a stated configuration is evaluated, which is legitimate for configurations related by a symmetry of the lattice and is why translation by one spacing is used as the control. Energies are reported in hartree and converted at 1 Ha = 27211 meV.

Model I: chain of atoms. $N = 7$ model atoms in a row, nuclear charge +1, one valence electron each, wrapped labelling so that translation by one spacing is an exact relabelling. Two geometries were used: $r_c = 1.0\,a_0$ at spacings $a = 3.5$ to $8.0\,a_0$ for the polarisability and the density trend, and $r_c = 1.93\,a_0$ at $a = 5.5\,a_0$, $h = 0.448\,a_0$ for the corrugation, r_c in the second being fixed by the isolated atom's ionisation energy rather than chosen. Polarisability from the induced dipole at two field strengths with the partition frozen, at $N = 3, 5, 7$: $\alpha = 27.4N\,a_0^3$, flat to 1% per atom. Corrugation from $E(0)$ and $E(\frac{1}{2})$; on the $a = 5.5$ chain, $E(0) = -1.631277$ Ha throughout, with $E(\frac{1}{2}) = -1.839$ Ha under own-core exclusion (808 meV per electron) and -1.617 Ha under exclusion from every core (54 meV). These bracket the answer; the electron-volt scale quoted in §3 is the own-core end, and the ~0.8–1.5 eV range across spacings from 3.5 to $8.0\,a_0$ is measured on the $r_c = 1.0$ chain.

Model II: cable of ions surrounded by electrons. $N = 16$ sites at $a = 4.5\,a_0$ on the axis, kernel charge +3, one core domain per site carrying two electrons and confined within $c_r = 2.4\,a_0$ of the axis, one valence electron per site outside that radius: 32 domains, neutral, $r_c = 0$ with the nuclear singularity clamped at an absolute $0.5\,a_0$ so the potential is the same object at every mesh. Mesh 8 cells per spacing, $h = 0.5625\,a_0$, box $59.5\,a_0$, grid 106^3. Only the shell slides. Energies at $d = \frac{1}{2}$ with M middle carriers displaced, against the undisplaced reference $E = -160.133661$ Ha:

M	4	8	12
E (Ha)	−160.352807	−160.312817	−160.281810
ΔE (meV)	−5963	−4875	−4031

Equation (3) on $(4,8)$ gives $\varepsilon_{\text{bulk}} = 272$ meV and $\varepsilon_{\text{junc}} = -3526$ meV per junction; on $(8,12)$ it gives 211 meV. The linearity test predicts $\Delta E(12) = -3787$ meV against -4031 measured, a 6% miss, and the spread between the two estimates is quoted as the uncertainty: 240 ± 30 meV. The $(4,8)$ pair is the cleaner, both its junctions lying well inside the chain, where at $M = 12$ only two sites remain on either side.

Stiffened carriers. Identical geometry, mesh and ion lattice, with the coefficient of $|\nabla\psi_i|^2$ raised from $\frac{1}{2}$ to 1 for the shell domains only, the cores keeping $\frac{1}{2}$. The coefficient enters three places — the imaginary-time step, the $H\psi$ diagnostic and the energy sum — and all three must carry it or the reported energy does not belong to the state produced. The explicit step requires the diffusion number below 1/6, so the timestep is halved in proportion; the stiffened runs are correspondingly less relaxed. $\Delta E(4) = -8447$ meV, $\Delta E(8) = -8352$ meV, giving $\varepsilon_{\text{bulk}} = 24$ meV per carrier.

The rate estimate of §6. Equation (4) evaluated at $T = 300$ K ($kT = 25.9$ meV) with $a = 4.5\,a_0 = 2.381$ Å, $n = 1/a^3 = 7.4 \times 10^{28}$ m^{-3}, $M = 28\,u$, and $K = 2\pi^2 V_0/a^2$ from the assumed sinusoid, giving $\nu = (1/a)\sqrt{V_0/2M}$:

V_0 (meV)	K (eV Å^{-2})	ν (Hz)	σ_0 (S/m)	$e^{-V_0/kT}$	σ (S/m)
1000	3.48	5.5×10^{12}	1.4×10^5	1.6×10^{-17}	2.3×10^{-12}
240	0.84	2.7×10^{12}	7.0×10^4	9.3×10^{-5}	6.5
24	0.084	8.5×10^{11}	2.2×10^4	0.395	8.8×10^3

Taking $M = m_e$ instead of an ionic mass raises ν to 6.1×10^{14} Hz and σ to 1.5×10^3 S m^{-1} on the middle row; that frequency is two orders above any lattice vibration, which is the argument for the ionic choice quite apart from the physical one. The ± 30 meV uncertainty on the middle row is a factor 3.2 in σ.

Controls applied. Translation by one full period must return the same energy: satisfied to nine digits ($E(0) = E(N)$ exactly) on the wrapped geometries. Repeat runs of one configuration agree to nine digits, so the solver is deterministic and any discrepancy is attributable to the model or the input. Domains related by a symmetry must have equal eigenvalues: this is what exposed the terminal domains of a finite cable binding an electron-volt harder than the interior, and hence the necessity of (3). Euler–Lagrange residuals are 4×10^{-3} for the chain and 2–3×10^{-2} for the cable.

References

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